

Social Media Engagement Analytics Using Python

Project Objective

- To analyze social media engagement patterns using Python.
- To understand how likes, comments, shares, impressions, and engagement rates vary across users and content types.
- To identify the best-performing post types, categories, hashtags, and sentiments.
- To analyze audience behavior based on age, device type, country, and account verification status.
- To generate business insights and recommendations for improving social media performance.

Problem Statement

- Social media platforms generate massive volumes of engagement data—likes, comments, shares, impressions, watch time, and more. Analyzing such data helps companies understand user behavior, identify trends, and improve content performance.
- In this assignment, you will work with a social media dataset containing social media engagement metrics. The task involves data cleaning, transformation, NumPy/Pandas operations, exploratory data analysis, visualizations, and generating insights.

Dataset Description

- Dataset Name: Social Media Engagement Dataset
- Total Records: 5000 rows
- Dataset Type: Structured CSV dataset
- Domain: Social Media Analytics
- Source: GitHub Repository

Key Features in Dataset

- User demographics (age, gender, country)
- Post details (post type, category, hashtags)
- Engagement metrics (likes, comments, shares, impressions)
- Watch time and follower count
- Sentiment analysis
- Device type and verification status

Dataset Columns

Column	Description
user_id	Unique identification number assigned to each user
age	Age of the user
gender	Gender of the user
country	Country of the user
post_id	Unique identification number assigned to each post
post_type	Type of content posted (video, image, reel, text, etc.)

post_category	Category/topic of the post (food, fitness, fashion, travel, etc.)
likes	Total number of likes received by the post
comments	Total number of comments received by the post
shares	Total number of shares received by the post
watch_time_sec	Total watch time of the content in seconds
impression_count	Number of times the post was viewed or displayed
posted_at	Date and time when the post was published
follower_count	Total number of followers for the user account
is_verified	Indicates whether the account is verified or not
device_type	Device used to access/post content (mobile, desktop, tablet)
sentiment	Sentiment label associated with the post (positive, negative, neutral)
hashtags	Hashtags used in the post
engagement_rate	Overall engagement performance calculated from interactions

Data Cleaning Process

Missing Value Handling

Checked missing values using:

- isnull()
- isna()

Numerical Columns

- Missing values were filled using median because median is less affected by outliers.

```
for col in numeric_cols:
    social_df[col].fillna(social_df[col].median(), inplace=True)
```

Categorical Columns

- Missing values were filled using mode because mode represents the most frequent category.

```
for col in categorical_cols:
    social_df[col].fillna(social_df[col].mode()[0], inplace=True)
```

Duplicate Handling

- Checked duplicate rows using: duplicated()
- Removed duplicate records to avoid repeated analysis.

Data Type Conversion

- Converted posted_at column into datetime format using: pd.to_datetime()

Category Standardization

- Standardized categorical values such as:
 - gender labels
 - sentiment labels
 - post types

Outlier Analysis

Outlier Detection Methods

- Boxplot visualization
- Skewness analysis
- Kurtosis analysis Columns with Outliers
- engagement_rate showed significant outliers.

Why Outliers Existed

- Some social media posts became viral and generated extremely high engagement.
- Viral content naturally creates unusually high engagement rates.
- These outliers represented real business behavior rather than incorrect data.

Log Transformation

- To reduce extreme positive skewness in engagement_rate.
- To compress very large engagement values.
- To improve visualization and statistical interpretation.
- To make the data distribution more stable and manageable.

Impact of Log Transformation

- Reduced skewness significantly.
- Improved distribution balance.
- Reduced the influence of extreme values.
- Improved readability of plots and statistical analysis.

Why Outliers Still Existed

- Some viral engagement values remained even after transformation because they represented genuine audience behavior.

Findings

- Video content generated higher engagement and impressions compared to other post types.
- Food-related content received the highest average likes.
- Brazil showed the highest engagement rate among all countries.
- Users around the age of 40 were more active in the dataset.
- Verified accounts achieved slightly better engagement than non-verified accounts.
- Mobile users recorded the highest average watch time.

- Negative sentiment posts generated the highest likes, comments, shares, and engagement.
- Hashtags such as #lifestyle, #tech and #travel generated the highest average engagement rates, indicating stronger audience interest in these topics.
- Log transformation helped reduce skewness and improve analysis of engagement-related data.

Recommendations

- Focus more on video-based content strategies.
- Optimize content for mobile users.
- Use high-performing hashtags to improve reach.
- Analyze emotional audience behavior for content planning.
- Increase engagement strategies for non-verified accounts.
- Use audience-focused posting strategies for better impressions and engagement.

Conclusion

- This project successfully analyzed social media engagement behavior using Python-based data analytics techniques.
- The analysis identified important engagement patterns related to post type, sentiment, device usage, demographics, and content categories.
- Video content, mobile usage, and emotional sentiment emerged as major drivers of engagement performance.
- The project demonstrated how data cleaning, statistical analysis, visualization, and business insights can be used together to support social media decision-making and audience engagement strategies.